Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | McLean, VA (open to SF / NYC)

PROFESSIONAL SUMMARY

AI agent engineer who **builds and ships enterprise AI agents** - Sierra's core product. Built AgentProbe end-to-end: a **3-agent LangGraph platform** (Orchestrator, RAG Tracer, Auto Fix) that handles multi-step reasoning, tool calls, and inter-agent handoffs across task domains - the same architecture Sierra deploys for enterprise customers in retail, healthcare, and fintech. **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates - agents are reliable in production before they touch customers. **Healthcare AI**: built AI on FHIR - HIPAA-compliant LLM + FHIR API integration with ~60% retrieval time reduction. **8+ years Python production** at Entrata (100K+ MAU, 1.6M records, 120+ Agile sprints). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Multi-Agent Systems (LangGraph, Tool Calls) | Enterprise Agent Deployment | LLM Evaluation & Reliability Frameworks | Python (8+ years production) | RAG Pipelines & Vector Search (ChromaDB) | Healthcare AI (FHIR, HIPAA Compliance) | FastAPI / REST Platform Layer | Enterprise-Scale Delivery (100K+ MAU) | Prompt Engineering & LLM Routing | React / TypeScript / GraphQL | AWS / Docker / CI/CD | Cross-Functional Technical Communication

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Served as **primary technical owner for cross-functional enterprise product delivery** across **20+ full-stack modules** (Python, TypeScript, React, PostgreSQL, GraphQL) - shipped production systems serving **100K+ monthly active users** across **120+ Agile sprints**, partnering directly with product, engineering, and business stakeholders.
- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - delivered production AI on regulated enterprise data with **100% PHI compliance, zero security incidents** using Libsodium-encrypted AWS S3 pipelines.
- Led Docker-based containerization for 30%+ of microservices, cutting execution time by **50%**; improved production query performance by **30%**; maintained **99.5% uptime** across enterprise SaaS; built 5+ debugging platform tools adopted across squads, reducing resolution time by **40%**.
- Eliminated 80% of manual operational workload by designing and automating decision-support workflows in Python and PL/SQL.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Translated complex technical concepts for 30+ students and designed supplementary Python exercises - the same communication pattern needed when shipping agents for enterprise customers across diverse industries.

PROJECTS

AgentProbe: Enterprise AI Agent Platform - Built to Ship, Not Demo OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · TypeScript · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the architecture Sierra deploys for enterprise customers - production AI agents end-to-end. **Agent architecture**: 3-agent LangGraph system (Orchestrator routes tasks across domains, RAG Tracer validates retrieval quality, Auto Fix closes hallucination loops) with tool calls, failure handling, and inter-agent state handoffs - runs across retail, healthcare, and fintech task patterns. **Reliability framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD gates - agents don't reach enterprise customers until they pass the quality bar. **LLM stack**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama with dynamic selection; ChromaDB vector store + embeddings + retrieval scoring. **Platform**: 15-endpoint FastAPI backend + React observability dashboard.
github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: Healthcare Enterprise AI - HIPAA-Compliant LLMs + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · GraphQL

End-to-end AI agent integration for healthcare - one of Sierra's target verticals. NLP routing natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC. ~60% retrieval time reduction.** Demonstrates production-grade AI for regulated enterprise customers - the pattern Sierra's healthcare clients require. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, Tool Calls, LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering, Context Engineering

Frontend & Backend: React, FastAPI, Flask, GraphQL, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum, Claude Code, GitHub Copilot, Cursor